OP-AMP: INVERTING AND NON-INVERTING

1. Operational Amplifier (OP-AMP)

An operational amplifier is a high-gain electronic amplifier that amplifies the difference between two input voltages. It has two input terminals, one output terminal, and power supply terminals.

The two input terminals are:

- Inverting terminal (−)
- Non-inverting terminal (+)

An op-amp can be used to perform amplification, addition, subtraction, and other mathematical operations.

2. Inverting Amplifier

Definition: An inverting amplifier is an op-amp circuit in which the input signal is applied to the inverting terminal (−), while the non-inverting terminal (+) is connected to ground.

Working:

1. The input signal is applied through an input resistor R_{in} to the inverting terminal.
2. A feedback resistor R_f is connected between the output and the inverting terminal.
3. The non-inverting terminal is grounded.
4. The output signal is amplified and has a phase difference of 180° from the input signal.

Voltage Gain Formula:

$$ A_v=\frac{V_{out}}{V_{in}}=-\frac{R_f}{R_{in}} $$

Where:

- A_v = Voltage gain
- V_{out} = Output voltage
- V_{in} = Input voltage
- R_f = Feedback resistance
- R_{in} = Input resistance

The negative sign indicates that the output signal is inverted.

Characteristics:

1. Output is 180° out of phase with the input.
2. Voltage gain depends on the ratio of feedback resistance to input resistance.
3. Uses negative feedback.
4. Input resistance is approximately equal to R_{in}.

Applications:

- Signal amplification
- Audio circuits

- Signal processing
- Weighted summing circuits

3. Non-Inverting Amplifier

Definition: A non-inverting amplifier is an op-amp circuit in which the input signal is applied to the non-inverting terminal (+), while the inverting terminal (−) is connected to a feedback network.

Working:

1. The input signal is directly applied to the non-inverting terminal.
2. A feedback resistor \(R_f\) is connected between the output and the inverting terminal.
3. Another resistor \(R_1\) is connected between the inverting terminal and ground.
4. Negative feedback controls the voltage gain.
5. The output signal is amplified without reversing the phase of the input signal.

Voltage Gain Formula:

\[A_v=\frac{V_{out}}{V_{in}}=1+\frac{R_f}{R_1} \]

Where:

- \(A_v\) = Voltage gain
- \(V_{out}\) = Output voltage
- \(V_{in}\) = Input voltage
- \(R_f\) = Feedback resistance
- \(R_1\) = Resistance connected to ground

Characteristics:

1. Output is in phase with the input.
2. Voltage gain is equal to or greater than 1 in the standard configuration.
3. Has very high input resistance.
4. Uses negative feedback.
5. Does not invert the input signal.

Applications:

- Voltage amplification
- Buffer circuits
- Sensor signal amplification
- Audio systems

4. Difference Between Inverting and Non-Inverting Amplifiers

Inverting Amplifier	**Non-Inverting Amplifier**
Input is applied to the (−) terminal.	Input is applied to the (+) terminal.
Output is 180° out of phase.	Output is in phase with the input.
Gain: $-R_f/R_{in}$	Gain: $1+R_f/R_1$
Input resistance is approximately R_{in}.	Input resistance is very high.
Voltage gain magnitude can be less than or greater than 1.	Standard voltage gain is at least 1.
Used in signal inversion and summing circuits.	Used in amplification and buffer circuits.

5. Conclusion

Inverting and non-inverting amplifiers are two basic op-amp configurations. An inverting amplifier produces an amplified output with a 180° phase shift, whereas a non-inverting amplifier produces an amplified output in the same phase as the input. The voltage gain of both circuits depends on the feedback resistor network.